Santiago del Palacio

Data Scientist & Astronomer

+46 070 013 7921
santiagodp1990@gmail.com
santiago-del-palacio
santimda
Gothenburg, Sweden

Professional Summary

Astrophysicist and data scientist with 10+ years of experience in statistical and physical modelling, proficient at addressing complex research problems with analytical and computational tools. Skilled programmer in applying machine learning methods, extracting reliable conclusions from noisy and heterogeneous datasets, and translating quantitative findings into clear insights for technical and non-technical audiences to support decision-making. Experienced in collaborative international research environments, coordinating projects across teams and disciplines, and adapting quickly to new scientific and technical domains. Actively integrates AI-assisted tools into daily analytical and development workflows.
Author of 8 first-author and 60+ co-authored peer-reviewed publications (including Nature and Science Advances); Principal Investigator on 18 and Co-Investigator on 26+ competitively awarded international research projects involving major observatories; delivered 18+ conference presentations internationally.
Currently seeking to apply strong analytical and statistical skills to roles with direct societal impact, particularly in health data, register-based research, or public sector analytics.

Work Experience

03/2022 – 04/2026 **Postdoctoral Research Fellow**, *Chalmers University of Technology, Dept. of Space, Earth and Environment*, Gothenburg, Sweden

- Led an independent, cross-disciplinary research programme, defining scientific direction and coordinating collaborations across multiple institutions. This highly competitive fellowship also required to work towards bridging multiple research groups within the division.
- Developed and validated physical models using statistical inference and numerical methods; quantified parameter uncertainties using Bayesian and frequentist approaches; developed tools for interpreting multi-wavelength observational datasets (radio, mm-wave, X-rays).
- Co-supervised 1 Ph.D. student, providing guidance in research design, data analysis, and problem-solving.
- Organised weekly departmental seminars to improve engagement and knowledge sharing within the division.
- Served as referee for major journals and as member of international time-allocation committees for ALMA and *XMM-Newton*, evaluating project quality, technical feasibility and scientific impact, and providing constructive and actionable feedback.

04/2019 – 03/2022 **Postdoctoral Research Fellow (CONICET)**, *Argentine Institute of Radioastronomy*, Buenos Aires, Argentina

- Led an independent research programme, including project design, development of theoretical and numerical models, and coordinating international collaborations and multi-site observational campaigns.
- Built automated data processing pipelines in Python for pulsar monitoring within the PuMA Collaboration.
- Supervised 3 M.Sc. students, defining project scope, work strategy, and continuous mentoring.

03/2014 – 04/2019 **Ph.D. Research Fellow (CONICET)**, *National University of La Plata*, La Plata, Argentina
- Developed numerical radiation-transport codes (FORTRAN) to model non-thermal emission in astrophysical systems.
- Designed and executed multi-observatory campaigns; analysed and interpreted data from radio, X-ray, and gamma-ray facilities.
- Awarded the *Carlos M. Varsavsky Prize* for best Ph.D. thesis in astronomy in Argentina (2018–2019 cohort).

02/2011 – 03/2022 **Teaching Assistant / Head of Practical Assignments**, *National University of La Plata*, La Plata, Argentina
Taught several courses on Calculus, Algebra, and Astrophysics at undergraduate and graduate level for both science and engineering students. Student evaluations consistently remarked my clarity at explaining complex concepts and good predisposition overall.

Education

2014 – 2018 **Ph.D. in Astronomy**, *National University of La Plata*, La Plata, Argentina, Thesis mark: 10/10
Thesis: *Non-thermal emission from systems harbouring massive stars.*
Award: *Carlos M. Varsavsky Prize* (best Ph.D. thesis in astronomy in Argentina, 2018–2019).

2008 – 2014 **Licenciatura in Astronomy (equiv. B.Sc. + M.Sc.)**, *National University of La Plata*, La Plata, Argentina, GPA: 9.56/10
Award: *Joaquín V. González Prize* (distinction for highest-achieving students, La Plata City Council).

Technical Skills

Programming Python (advanced: NumPy, SciPy, Astropy, scikit-learn, Matplotlib, Pandas), FORTRAN (advanced), SQL (proficient), Bash (intermediate), R (basic)

Tools & Platforms Git/GitHub, Linux/Unix, LaTeX, Jupyter, BigQuery, Looker Studio, LaTeX; scientific data-analysis pipelines; AI-assisted development (GitHub Copilot, LLMs)

Methods Numerical simulations, statistical modelling, uncertainty quantification, Bayesian inference, causal inference, numerical methods, data pipeline design, data visualisation, scientific image analysis, time-series analysis, signal processing, machine learning, dashboards

Soft Skills Project management, scientific writing, public communication, international and interdisciplinary collaboration, student supervision

Languages

Spanish Mother tongue

English Fluent (C2)

Swedish Upper Intermediate (B2/C1)

Outreach & Science Communication

2015 – 2026 14 public outreach activities, including public talks on astronomy and astrophysics (e.g. *Neutron Stars and Pulsars*, Slottsskogsobservatoriet, Gothenburg, 2026), lectures at the Swedish National Space School *Rymdforskarskolan* (2025), guided visits at open observatory nights, and co-authoring press releases.

References

Colleagues and line managers have highlighted my independence, contribution to a positive work environment, and capacity to move across different fields. References available upon request.